
Impacts of COVID-19 on Buildings Energy Usage

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Source: New York Times [1]

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1. Abstract

Buildings in the United States consume 70% of electricity energy. Energy consumption varies greatly between buildings and can be influenced by several factors. Energy usage varies greatly in educational and office buildings, while multifamily buildings are more consistent. Stay-at-home orders in response to COVID-19 shifted the workforce away from office buildings, increasing time spent in residential buildings. This study analyzes energy consumption changes relating to COVID-19. This study contributes towards the discussion of a general shift towards work from home settings and how work-from-home programs could contribute to reducing greenhouse gasses and mitigating climate change.

Visual Abstract:

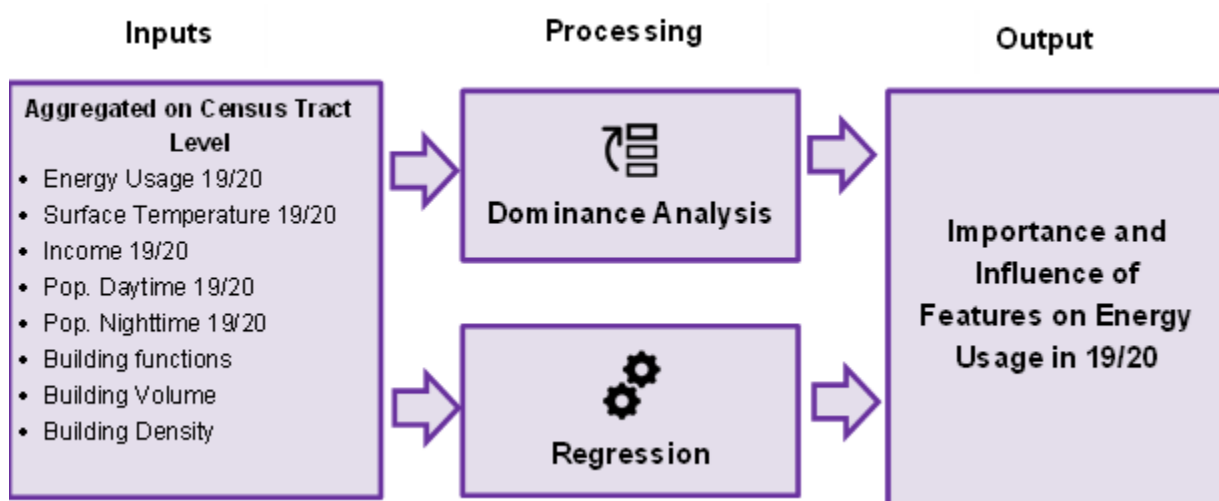


Figure 1 Visual Abstract

Keywords

COVID19, Energy Usage, Feature Importance, ElasticNet Regression

2. Introduction

2.1 Research on Energy consumption before and after COVID19

2.1.1 Building energy consumption and its impacts

Buildings in the United States consume 70% of electricity energy (with commercial and residential property types made up 21% and 18% respectively) [2]. Many factors influence building energy consumption, including ambient temperature, building characteristics, and the performance of facilities such as lighting, heating, ventilating, and air conditioning systems [3] [4] [5] etc. Building energy consumption is also increased significantly by urbanization, climate change, environmental policies, etc., posing challenges for sustainable development. Challenges include increasing greenhouse gas emissions, exacerbating air pollution, and interrupting reliable energy supply [6]. Consequently, insight into building energy consumption is important to the sustainability of urban development.

2.1.2 Building energy under the impact of Covid-19

COVID-19 affected working, living, and traveling behavior, which subsequently impacted energy consumption. There have already been a number of studies focusing on environmental and energy changes based on COVID-19. Elavarasan et al. and Akrofi et al. [7] investigated the changes in power load caused by COVID-19, and the problems associated with it were analyzed. Eirini et al. [8] optimized the hybrid working model and activity-based workspace environments to shrink the energy demand in the office building sector up to 50% in comparison to the pre-pandemic situation. Syed et al. [9] analyzed the actions taken by governments towards building energy consumers and evaluated the feasibility of governmental-wise action plans to elevate the energy shortage scenario.

Nevertheless, the study of building energy consumption change related to COVID-19 remains scarce. In this study, we are focusing on how building energy consumption was affected by stay-at-home orders in New York City.

2.2 Project Objectives

Previous research and reports mostly hint towards the relationship of single parameters to the change of Energy Usage in 2019 and 2020. In our study we aim to provide a wider perspective in the relationship between a selection of influencing factors on the Energy Usage before and after COVID19. The results can be seen as a starting point for discussions related to the potential benefits remote work entails towards lowering general energy consumption of buildings.

3. Data

3.1 2019/2020 Land Surface Temperature

This analysis used satellite Data obtained from Landsat 8 via Google Earth Engine. [10] Using only non-corrupted satellite images with <10% cloudiness we took the average temperature from two months per year. Due to data quality constraints, only images from March and June 2019 and January and July 2020 were used.

3.2 Income

Income was obtained through Census Data in 2019 and 2020.using the R-package tidyCensus. [11].

3.3 Landscan Data

Landscan Data [12] was used to extract day and night population for New York City in 2019 and 2020. The spatial resolution for the population is 100m². By comparing the difference of day and night population we can determine the temporal patterns of the population density. The population is further aggregated to zip code level to combine with other features to do the regression analysis.

3.4 Building functions

In our study we include the portion of Residential vs Commercial vs other building functions. The base of the data is Primary Land Use Tax Lot Output (PLUTO) obtained through the open data portal of NYC [13]. The Building Class for each Tax zone is categorized into Commercial, Residential and Other Building functions. Finally, the portion of each building function group is summarized on Census Tract Level. The exact determination of the Categorization can be seen in Table 5 in the appendix.

3.5 Building Height and Footprints

Building height data is sourced through the DOITT Footprint data which contains the average roof height of each building [14]. This data is standardized using Census Tract Level to obtain the average Building density and average building volume per Census Tract.

3.6 Energy Data

Building energy data excludes observations where the identifier (BBL number - Borough, Block, and Lot) was duplicate, missing, or where the BBL number was in the wrong format. Energy values for included buildings are based on “Site EUI (kBtu/ft²)”, a measurement that calculates the energy use of the site divided by the area of the respective property [15], [16], [17].

3.7 Census Tracts

Census Tracts dictate the geometry over which we aggregate all of the gathered data. Census Tracts were selected because they provide an arguably regular grid over the city’s extents, while providing a reasonable association towards the granularity of other publicly available data [18].

4. Methodology

4.2 Energy Consumption Change Analysis

In this section, we explore whether the pandemic has had a significant impact on electricity use in different types of buildings. The three building types with the largest share of the Energy and Water Data Disclosure for Local Law 84 dataset are Multifamily Housing, Office, and K-12 School in order, while the number of other building types is minimal. As a result, the analysis in this section focuses only on these three types of buildings. We merged the data from 2018 to 2020 [17] [15] [16] excluding buildings without a complete three years of building energy consumption data. Figure 9 in the appendix shows the location of these three types of buildings.

We defined 2019 as the year before the COVID-19 impact and 2020 as the year affected by the epidemic. We integrated the Site Energy Use (kBtu) of these three types of buildings to the census tracts level and visualized the change in energy consumption of these three types of buildings at the census tracts level between the two years. Figure 3 shows that the total energy consumption of most census tracts in these three types of buildings has decreased in varying degrees.

We analyzed the distribution of Site EUI (kBtu/ft²) at the building level in 2019 and 2020 for each of the three types of buildings. This change of distributions is visualized in Figure 2. The average values of all three building types have decreased, with the decrease in building energy consumption in Office and K-12 schools being more drastic than Multifamily Housing (Table 4 in appendix shows the detailed statistics description of those distributions).

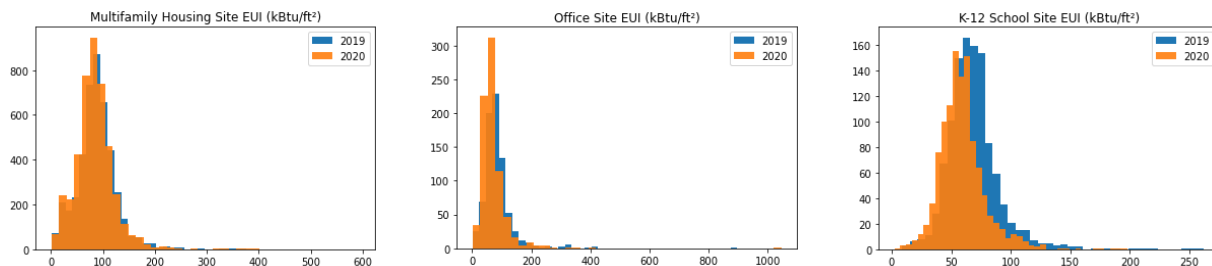


Figure 2 Histograms showing the change of Energy Usage for Multifamily Housing, Offices, and Educational Buildings

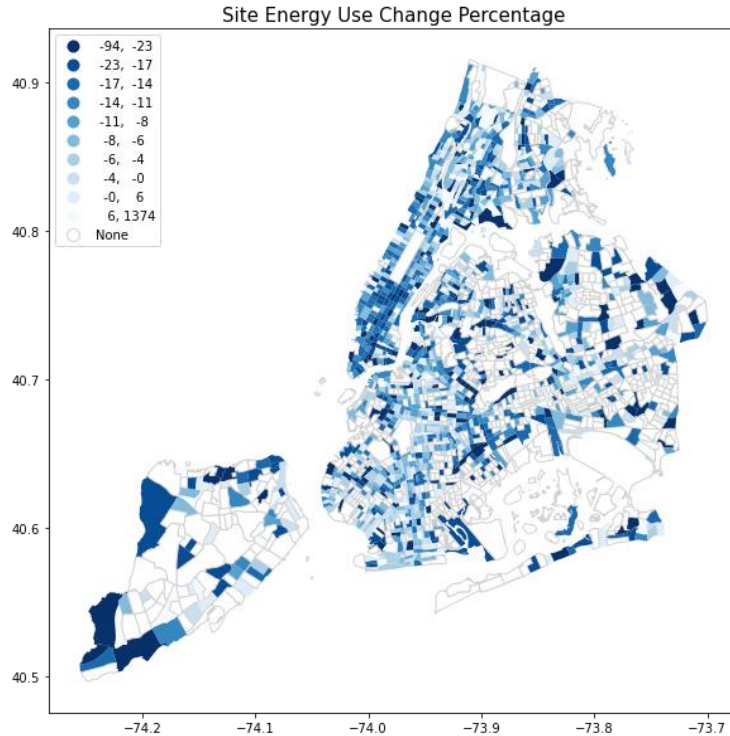


Figure 3 Site Energy Use Change Percentage per Census Block

Since a general trend of change in energy consumption is possible, we normalized the data to confirm that the decrease in energy consumption is due to the impact of the pandemic. Specifically, we divide the Site EUI of each building in 2019 (kBtu/ft²) by the Site EUI of 2018 (kBtu/ft²) as the pre-pandemic ratio, and the Site EUI (kBtu/ft²) in 2020 divided by Site EUI (kBtu/ft²) in 2019 for each building as the 2020 ratio. We then propose the null hypothesis that the average/distribution of the pre-pandemic ratio and the 2020 ratio for all three types of buildings are the same.

Table 1 Test Results Change of Energy Consumption for 2019 and 2020

	Multifamily Housing		Office		K-12 School	
	Ttest	Kstest	Ttest	Kstest	Ttest	Kstest
statistic	11.721	0.174	2.495	0.464	7.759	0.369
p-value	0.000	0.000	0.012	0.000	0.000	0.000

For all three types of buildings, we can reject the null hypothesis that the average/distribution of building energy consumption change ratio in 2019 and 2020 is the same at a 95% confidence level.

4.3 Feature Ranking

4.3.1 Correlation

As a first step towards establishing a ranking of influence on energy usage, a correlation between the used variables is established. Within the parameters that we use multicollinearity is found between various fields which poses a special challenge towards applying Regression models on our Census Tract level data [19].

In both correlation matrices (Figure 4 and Figure 5) there are significant correlations between Building Density and Building Volume as well between Residential Buildings and Commercial Buildings. For the latter the relationship is linear leading to an obvious structural correlation in the data. Additionally, for both years the correlation between the calculated population during day vs the Building Volume is extremely high. Special attention needs to be given to the correlation between the parameters for 2020 towards the Energy use in 2020, the correlation is for all variables relatively low, posing the challenge that a linear regression might deliver low statistical significance for each coefficient.

	BldgVolume	BldgDensity	Commercial	Others	Residential	income2019	PopDay19	PopNight19	Temp19Mean	EnergyUse_kWh19
BldgVolume										
BldgDensity	0.88									
Commercial	0.57	0.60								
Others	0.27	0.26	0.35							
Residential	-0.55	-0.57	-0.90	-0.72						
income2019	0.47	0.44	0.14	0.09	-0.14					
PopDay19	0.87	0.81	0.62	0.18	-0.54	0.38				
PopNight19	0.20	0.12	-0.05	0.15	-0.03	0.01	0.05			
Temp19Mean	-0.15	-0.13	0.02	-0.18	0.07	-0.16	-0.05	-0.17		
EnergyUse_kWh19	0.78	0.75	0.53	0.31	-0.54	0.32	0.75	0.17	-0.13	

Figure 4 Correlations for 2019

	BldgVolume	BldgDensity	Commercial	Others	Residential	income2020	PopDay20	PopNight20	Temp20Mean	EnergyUse_kWh20
BldgVolume										
BldgDensity	0.88									
Commercial	0.57	0.60								
Others	0.27	0.26	0.35							
Residential	-0.55	-0.57	-0.90	-0.72						
income2020	0.44	0.42	0.12	0.09	-0.13					
PopDay20	0.87	0.81	0.62	0.18	-0.54	0.35				
PopNight20	0.20	0.12	-0.05	0.15	-0.03	0.01	0.05			
Temp20Mean	-0.25	-0.32	-0.08	-0.17	0.14	-0.25	-0.16	-0.24		
EnergyUse_kWh20	0.74	0.71	0.50	0.32	-0.52	0.27	0.68	0.20	-0.23	

Figure 5 Correlations for 2020

4.3.2 Dominance Analysis

A Dominance Analysis is used to determine the relative importance of predictors for the regression model. The principle is laid out in determining the dominance of a regressor over another by comparing their contribution to the R^2 factor¹.

In other words, in this analysis, we decompose R^2 into the proportion explained by each regressor. In practice the source code we run is calculating a linear regression for every possible combination of regressors, in our case $2^p - 1 = 2,047$ combinations with $p = \text{number of regressors}$. The contribution of each regressor is measured by the increase in R^2 resulting from adding the regressor to each regression model.

In our implementation we make use of the package “dominance-analysis” [20] to calculate R^2 of linear regressions using the permutations of the regressors. The package is also used to visualize the contribution of each regressor. The importance of each regressor for 2019 and 2020 is represented in Figure 6, which shows the contributions to R^2 in the form of a pie chart.

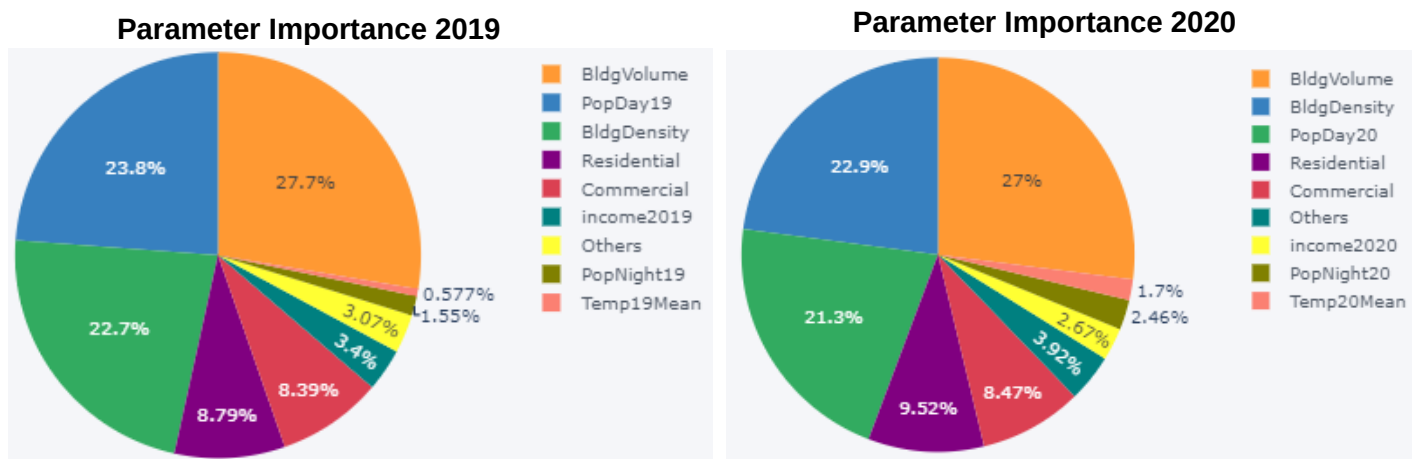


Figure 6 Parameter Importance on R^2 calculated through Dominance Analysis

The most influential parameter is the building volume, followed by building density while the least important parameter for predicting energy data is the mean surface temperature per census tract. A similar pattern can be seen in 2020 with other building types being interchanged with Income. What this Dominance Analysis doesn't deliver is the actual numerical relationship in the form of a coefficient to the dependent variable, it is one goal of this study to see the influence of each regressor to the actual result. Therefore, we also run a regression model while knowing what the most influencing parameters are from the Dominance Analysis.

4.3.3 Elastic Net Regression

For the implementation of the regression models, we made the baseline decisions to compute the regressions on standardized input data as well as omitting the calculation of an intercept.

¹ R^2 is the proportion of variation in a dependent variable that can be attributed to independent variables (Regressors)

This limits the impact of high variances of certain input variables and assumes that no energy is consumed if all regressor's values are at zero in the respective census tract.

After the implementation and testing of Ridge and Lasso Regressions, we decided to apply Elastic Net as our Regression method. With R^2 being the same for each of these methods, we made this decision as regressions with Elasticnet allow a balance of both penalties from Lasso and Ridge Regressions, leading to a model that is based on coefficients selected and weighted according to their importance for predicting energy usage [21]. The implementation of ElasticNet involves the determination of a regularization parameter alpha, the choice of which is determined iteratively through the validation of a range of generated regression models using different alpha-values. The choice on the optimal alpha falls on the model which produces the highest R^2 , this process is visualized for both years in Figure 7. The chosen alpha for 2019 is 0.3933 and for 2020 is 0.0067. This indicates that for 2019 the coefficients have been determined with balanced Lasso and Ridge penalties. For 2020 the regression leans more towards penalties from the LASSO-method, meaning the coefficients are determined with sparse model weights.

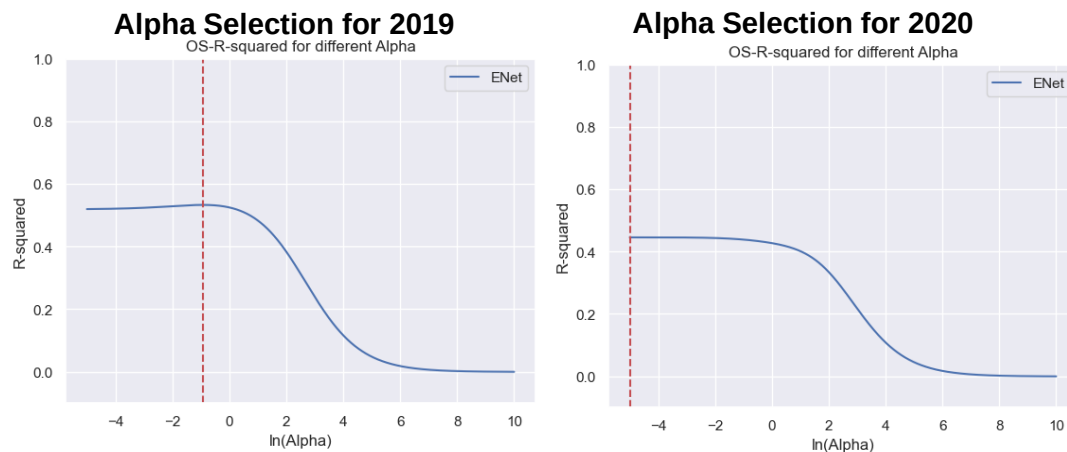


Figure 7 Regularization Parameter Alpha selection for ElasticNet Regression

5. Conclusions

From the statistical analysis of energy usage between 2019 and 2020 it can be concluded that there is significant change in energy usage in buildings. Most significant is the change of energy usage in offices and educational buildings.

This significant change of energy usage gives reason to investigate the influence of a selection of sociological, topological and physical parameters on energy usage in both years and how they changed through the invocation of stay-at-home orders.

By applying ElasticNet Regression, we obtain weights for each individual input variable.

The Regression is calculated for a separate parameter set for 2019 and 2020 respectively.

The ranking can be seen in Figure 8.

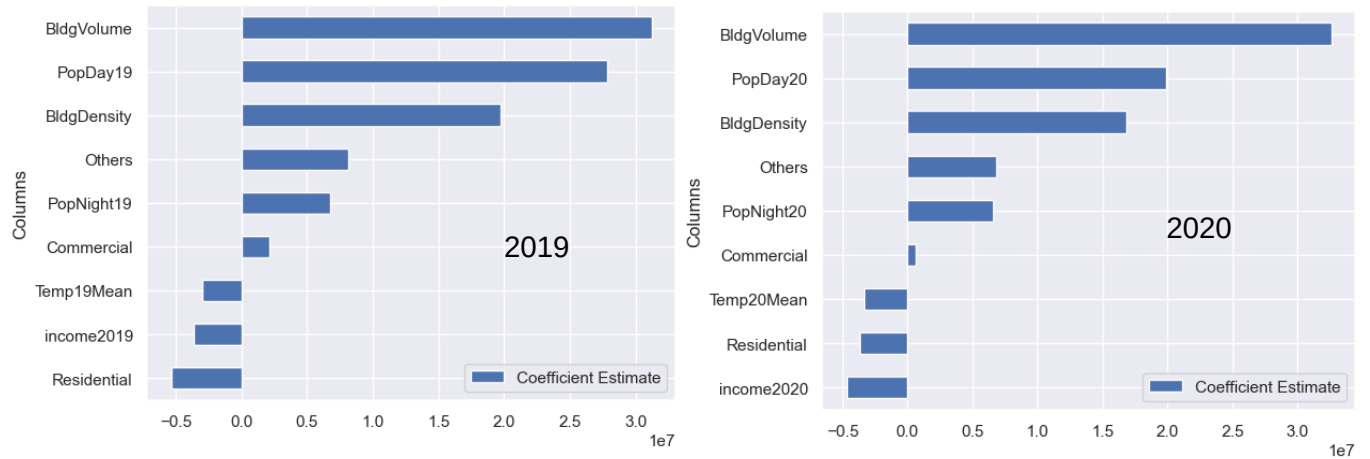


Figure 8 Ranking of Dependent Variables with Coefficients for 2019 and 2020

The in-sample R^2 for 2019 is 0.4626, for 2020 R^2 is 0.3958, meaning ca. 42% of the energy usage variability can be explained through the variables used as input to the regression model in both years. The ranking reveals that the biggest influence on energy usage is coming from the building volume and population present in the respective census tract during daytime in both years. The fact that the amount of buildings in a census tract drives the energy consumption is also represented through the high ranking of building density in each year. These 3 first parameters have also been classified as being the most important factors by far on the accuracy of the regression through the dominance analysis in section 4.3.2 Dominance Analysis.

The amount of other buildings such as universities, churches, and schools in the compared Census Tracts are ranked on 4th position in both years, albeit slightly less important in relation to Nighttime population in 2020, which stayed on a similar influence level in both years.

A significant change can be observed in the influence of the amount of commercial buildings, in 2020 the influence is lower which can be traced back to the fact that commercial buildings were less occupied in 2020 and therefore had a general lower share on the total building's energy usage in 2020 in NYC [22]. In both years surface temperature is ranked in 7th position. The negative relation can be explained through the fact that people tend to use more electricity during cold months than during summer [23]. Income is negatively correlated, Census Tracts with lower income tend to consume more electricity than Census Tracts with high income. Exploratory Data Analysis has shown that this is inversely the case for Manhattan, therefore this relation mostly applies towards Staten Island and gentrified census tracts in Brooklyn. The negative relation of the amount of residential buildings hints towards the fact that residential buildings tend to use significantly less electricity than commercial or other building types.

The most significant takeaway from this analysis is that the influence of commercial buildings on electricity has been reduced significantly through the COVID-19 orders, this study therefore proves with statistical methods that work-from-home concepts might contribute to a reduction of total energy usage of buildings, and therefore a reduction of Greenhouse gasses, leading to a further mitigation of climate change.

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7. Appendix

Table 2 Team Member Contribution

Team Member	Contribution
Matthias Fitzky	Team Management Data Gathering/Processing Dominance Analysis, Regression, Conclusions Presentation
Yanchao Li	Data Gathering/Processing Building Energy Consumption Analysis Literatures Presentation
Taylor Gill	Report Formatting Presentation Proofreading
Tianqi Qian	Report Formatting Presentation
Qi Lin	Data Gathering/Processing Energy Consumption Change Analysis Presentation
Henry Cui	Report Formatting Presentation

Table 3 Summarization of PLUTO Types

PLUTO Code starting with	Summarized as
A, B, C, D, L, N, S	Residential
E, F, G, H, I, J, K, O, P, T, U, W, Y	Commercial
All remaining Types	Others

Table 4 statistics description of Site EUI distributions

	Multifamily Housing Site EUI (kBtu/ft ²)		Office Site EUI (kBtu/ft ²)		K-12 School Site EUI (kBtu/ft ²)	
Year	2019	2020	2019	2020	2019	2020
Mean	88.78	84.03	84.52	70.62	69.54	59.43
Standard deviation	41.42	38.61	55.74	53.25	24.67	20.26
25 percentile	67.70	63.70	58.82	47.70	55.98	47.30
50 percentile	87.20	82.10	75.75	60.75	66.85	57.20
75 percentile	106.80	101.70	96.85	80.07	77.93	67.22

Table 5 Explanation of Variable Names

Variable Name	Explanation
'BldgVolume'	Total Volume of Buildings per Census Tract calculated through Building Height and Building footprint area
'BldgDensity'	Building Density per Census Tract calculated through the Volume of Buildings divided by the Area of each Census Tract
'Commercial'	Portion of Commercial buildings in %
'Others'	Portion of Other buildings in %
'Residential'	Portion of residential buildings in %
'income2019'	Median Income per Census Tract in 2019
'income2020'	Median Income per Census Tract in 2020
'PopDay19'	Yearly average of Population present during daytime in 2019
'PopNight19'	Yearly average of Population present during nighttime in 2019
'PopDay20'	Yearly average of Population present during daytime in 2020
'PopNight20'	Yearly average of Population present during nighttime in 2020
'Temp19Mean'	Mean Surface Temperature over the full year of 2019 from Landsat 8
'Temp20Mean'	Mean Surface Temperature over the full year of 2020 from Landsat 8
'EnergyUse_kWh19'	Summarized Energy Usage per Census Tract in kilo watt hours in 2019
'EnergyUse_kWh20'	Summarized Energy Usage per Census Tract in kilo watt hours in 2020

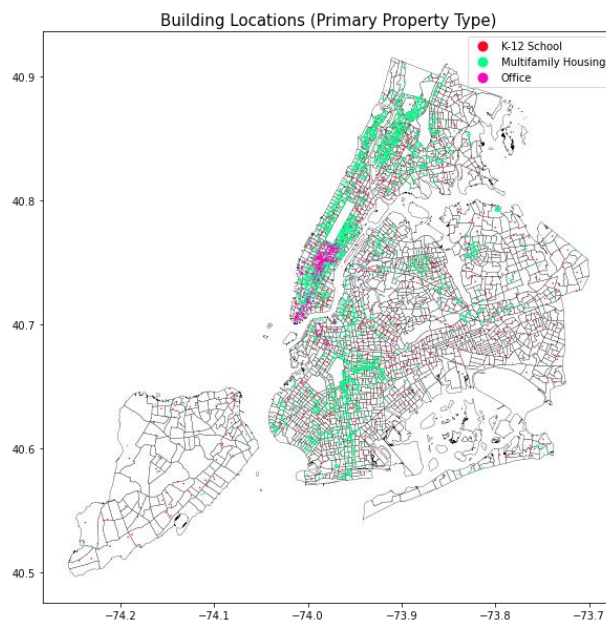


Figure 9 Building Locations